

**Black Mass in the European Union:
Classification and the Governance of Trade in Battery Scrap for Critical Minerals Recovery**

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September 2026

Abstract

Black mass is the powder produced when lithium-ion batteries are shredded. It contains the critical minerals that battery recycling is meant to recover, and it is at once waste, commodity, and strategic input. We theorize that the multiple identities of black mass produce governance problems of visibility, jurisdiction, and dependence. We treat EU battery circularity policy as an attempt at strategic coupling and black mass as the point where coupling holds or leaks. Empirically, we trace black mass through EU export declarations under the several customs codes that carry it, for 2022-2025, the first years in which battery scrap can be observed in trade data. Our findings show that black mass is visible only through fragmented and internally heterogeneous codes; that sorted material leaves the EU for refining destinations that the EU's waste rules reach only in part; and that EU exports depend on a few young and fluctuating corridors. While the EU has been a first mover in linking environmental governance, trade, and critical minerals policy, in this period its policies have not eliminated dependence on trade in black mass. Dependence appears in new places, including sorting industries inside the EU, the refining destinations outside it, and the classification boundary between them.

1. Introduction

Lithium-ion battery circularity has become central to climate, industrial, and critical minerals policy worldwide. The European Union (EU) now treats the recovery of battery materials not merely as waste management but as a pillar of strategic autonomy: the 2023 Batteries Regulation and the 2024 Critical Raw Materials Act both frame recycling as a route to securing domestic supply of lithium, nickel, cobalt, manganese, and graphite while reducing dependence on concentrated external suppliers. Underpinning this ambition is an implicit geography: circularity as a regional loop in which spent batteries are collected, processed domestically, and their recovered critical materials fed back into EU production.

Black mass is where that assumption is tested. It is the mineral-bearing intermediate powder produced when end-of-life batteries and production scrap are discharged, shredded, and mechanically separated; and it concentrates the recoverable lithium, nickel, cobalt, and manganese that make battery recycling strategically valuable. It is the point in the battery after-life at which recovered value is densest, and therefore the point at which that coupling either holds or leaks abroad. Yet black mass sits uneasily across three regulatory regimes. In waste law it is a residue, and from November 2026, when the 2025 amendment to the European List of Waste takes effect, a hazardous one; in recycling markets it is a traded feedstock; in critical minerals policy it is a strategic secondary resource. Because it is all three at once, it is difficult to classify, difficult to trace in trade statistics, and free to move through corridors well beyond the reach of the regulator setting recovery targets.

This tension motivates the paper's central research question: if battery circularity is meant to localize recovery and strengthen mineral security, why do EU-origin black-mass-related flows remain organized through concentrated hubs and extra-regional processing corridors? Put differently, what does the emerging trade structure of black-mass-related materials reveal about the governance challenges of battery circularity policy?

Answering this empirically has been difficult precisely because of the classification problem. The closest prior study, Gherasim and Michel (2024) documents that a large share of EU-origin black mass leaves the continent for South Korea and other Asian destinations, but its estimates rest on industry interviews and capacity maps because, harmonized classification and centralized statistics for black mass do not exist. This paper addresses that gap with EU export declarations for 2022-2025 from Eurostat's Comext database recorded under the Combined Nomenclature (CN), the EU's 8-digit product extension of the Harmonized System used in customs declarations worldwide. The analysis follows six CN codes that together approximate a black-mass-related trade ecology. That black mass can only be approximated in this way is the paper's first finding, not a limitation of its data. The paper then characterizes the structure of EU-origin flows through measures of concentration, corridor dependence, corridor persistence, and regionalization providing the first systematic trade-data evidence for a leakage pattern that prior work could establish only anecdotally.

The analysis is descriptive. It documents a starting point, the first four years in which battery scrap can be observed in trade data and asks whether trade structure in this period shows the concentration and extra-regional dependence that a global production networks (GPN) reading of EU circularity policy predicts. It does not estimate the effect of any regulation.

The findings show that it does, but unevenly. Trade in EU-origin black-mass and related materials is a layered system rather than a single diffuse market. The chemical mixture code is dispersed across many exporters and destinations; the battery scrap and refinery residue codes are concentrated on both sides, each organized through one or two near-exclusive bilateral corridors. Regionalization is partial: unsorted battery scrap stays inside the EU while sorted battery scrap and metal-bearing residues go to Singapore, South Korea, and Japan, where refining capacity is located, and Germany is the one member state where unsorted scrap goes in and sorted scrap comes out. In the battery scrap codes, the corridors that carry the trade are one to three years old. Battery circularity, in short, does not eliminate trade dependence; it reorganizes it across secondary-material hubs, processing corridors, and classification boundaries, producing three connected governance challenges (visibility, jurisdiction, and dependence) that the remainder of the paper develops.

The paper proceeds as follows. Section 2 develops the governance framework and situates black mass across waste, commodity, and critical-minerals regimes. Section 3 describes the data and methods. Section 4 presents the empirical results. Section 5 discusses the governance implications, and Section 6 concludes.

2. Policy background and theory: Battery circularity as a governance problem

2.1 Battery circularity as a trade-governance problem: A global production networks lens

Battery circularity should not be assumed to operate as a closed regional loop in which spent batteries are collected, processed, and returned to production entirely within a jurisdiction's borders. Reviewing critical circular economy scholarship, Kirchherr et al. (2025) discuss thermodynamic limits to full circularity and the risk that closed-loop narratives obscure the social and political dimensions of waste and resource flows. They emphasize that influential circular economy frameworks seek to minimize leakage rather than achieve perfect closure. The global circularity rate fell from 9.1% in 2018 to 7.2% in 2023, highlighting the gap between circular ambitions and material outcomes (Fraser et al., 2024, as cited in Kirchherr et al., 2025, p. 1972). These material limits are distinct from the question of geographical closure: materials can cross borders while remaining in productive circulation. This paper examines battery circularity as a contested and partial reorganization of material flows across technical, regulatory, and geopolitical systems.

The cross-border character of circularity is most visible in trade in waste itself. Towa et al. (2021) show that a region's measured circularity changes substantially once interregional flows of waste for treatment are counted: with trade included, Brussels' circularity rises from effectively

zero to 7.7%, because the region depends on recovery capacity located elsewhere. Circularity is thus a relational outcome organized through interregional flows of waste, treatment services, and recovered materials, not an autonomous regional achievement. Such secondary-material trade is also hub-concentrated: UNCTAD (2025) finds scrap copper flows dominated by a small set of exporters and importers led by the United States, Germany, and Japan. The concentration this paper observes in black-mass-related flows is therefore consistent with established secondary-material markets rather than an artefact of battery-specific data.

If circularity is relational, governing it requires a theory of how material flows, value, and dependence are organized across jurisdictions. The global production networks (GPN) framework offers this. Yeung and Coe (2015, p. 29) define GPNs as "organizational platforms through which actors in different regional and national economies compete and cooperate for a greater share of value creation, transformation, and capture through geographically dispersed economic activity." Three features are relevant here. First, GPN analysis foregrounds the uneven distribution of value capture across actors and territories. Second, it theorizes regional participation as strategic coupling, a dynamic process of coupling, decoupling, and recoupling in which a region's connection to a network can be deepened, redirected, or severed (Coe and Yeung, 2019). Third, it treats the state as an active network actor and "inter-scalar mediator" that can "facilitate, enhance, prolong or indeed end" the coupling of its territories to wider networks (Coe and Yeung, 2019, p. 782). The literature has also begun to extend to the "after-lives" of products and the recycling networks through which discarded materials re-enter production (Coe and Yeung, 2019, p. 778), precisely the flows that battery circularity creates.

Read through this lens, EU battery-circularity policy is an attempt at strategic coupling: a regional effort to capture the value embedded in secondary critical materials by building and securing domestic recovery capacity. The governance question is not simply whether Europe can recycle more, but where value and processing capacity concentrate within the resulting network, and whether the EU, as a supranational regulator, can steer flows that extend beyond its jurisdiction. Black mass is pivotal because it is the point in the after-life network where recoverable value concentrates and where coupling either succeeds or leaks abroad.

The GPN framework cannot answer every part of this question on its own; its authors describe it as "a necessary but not sufficient" tool for explaining regional development outcomes (Coe and Yeung, 2019, p. 793). The following sections therefore pair it with the regulatory and material specifics of black mass and the governance problems its trade structure generates.

2.2 From waste management to strategic coupling

If EU battery circularity policy is an attempt at strategic coupling, its regulatory record is where that attempt becomes legible: a shift from governing batteries only at the point of disposal to coupling their recovered value to the European economy. Directive 2006/66/EC governed batteries as waste, requiring collection, treatment, and recycling, and making producers pay for waste-

battery management (European Parliament and Council of the European Union, 2006). Who captured the recovered value was outside its concern.

That question now drives the agenda. Lithium-ion batteries are essential for secure energy transitions across electric vehicles, grid storage, and renewable integration (IEA, 2024a), so spent batteries are potential secondary sources of lithium, nickel, cobalt, manganese, and graphite: a waste-management challenge, but also an opportunity to recover strategic materials (Harper et al., 2019). In GPN terms, the secondary-materials node has become visible to the EU as a site of value capture.

Regulation (EU) 2023/1542 is the first binding EU instrument to operationalize this ambition at life-cycle scale. It replaces the Directive's waste focus with a framework spanning sustainability, performance, collection, recycling, repurposing, due diligence, and market access, tied explicitly to the Green Deal, competitiveness, circularity, and strategic autonomy. Its coupling instruments are mandatory recycled-content requirements for cobalt, lead, lithium, and nickel; material-recovery targets; and a digital battery passport for traceability (European Parliament and Council of the European Union, 2023). This is the state acting as the network actor and inter-scalar mediator described in Section 2.1.

The Critical Raw Materials Act, Regulation (EU) 2024/1252, makes the ambition quantitative. It designates lithium, cobalt, nickel, graphite, and manganese as strategic and sets 2030 benchmarks for Union capacity: at least 10% of annual consumption from extraction, 40% from processing, and 25% from recycling, with no more than 65% of any strategic raw material at any processing stage sourced from a single third country (European Parliament and Council of the European Union, 2024). Read as coupling targets, these figures state where the Union seeks to build capacity and cut dependence; circularity becomes a supply-security strategy as much as an environmental one. Analysts questioned this import-substitution logic from the outset, arguing that EU exposure arrives mainly through imported manufactured goods rather than raw materials (Le Mouél and Poitiers, 2023).

This widening of purpose produces the tension the framework anticipates. Circular economy scholarship centers on keeping materials in circulation rather than improving end-of-pipe disposal (Geissdoerfer et al., 2017; Kirchherr et al., 2017). Within the EU's critical minerals policy, the task becomes governing flows, building processing capacity, coordinating trade rules, and reducing dependence on concentrated supply: the work of a network actor rather than a waste regulator.

Coupling is not guaranteed by intent. Circular lithium-ion battery value chains require clarity over waste status, consistent transport rules, harmonized standards, and trade facilitation, since battery-waste trade may remain necessary where domestic capacity is insufficient (Mořsé and Rubínová, 2023). And circularity supports strategic autonomy only when processing and recycling capabilities exist within the governed value chain; otherwise loops remain transnational and value leaks to nodes the EU does not control (Jakimów et al., 2024).

Battery circularity is therefore a coupling problem, not a recycling problem alone: recycling reduces reliance on primary mining but can create new dependencies around collection, preprocessing, refining, trade corridors, and industrial hubs. Whether the Union's project holds or leaks turns on where recoverable value concentrates. In dominant recycling routes, that point is black mass, to which the next section turns.

2.3 Black mass between waste, commodity, and critical raw material

Black mass is a cross-regime object: a residue in waste law, a traded intermediate in recycling markets, a source of secondary supply in critical minerals policy. It occupies all three at once.

As waste, black mass is the product of pretreating end-of-life batteries and production scrap through discharge, dismantling, shredding, thermal treatment, and mechanical separation (Gherasim and Michel, 2024; Makuza et al., 2021), and it remains within waste-law categories even when destined for recovery. Alongside recoverable lithium, nickel, cobalt, and manganese, it may contain residual electrolytes, reactive lithium salts, fluorinated binders and solvents, graphite, copper, aluminum, and leachable metals (Niu et al., 2023; Zhang et al., 2025). The governing questions are whether it is legally waste, whether it is hazardous, whether it has reached end-of-waste status, and whether it may be shipped.

As commodity, black mass is traded and processed as secondary feedstock valued for its lithium, nickel, and cobalt and, depending on chemistry and route, manganese and graphite (Gherasim and Michel, 2024; Makuza et al., 2021). It is not standardized like refined lithium carbonate or nickel and cobalt sulfates: despite emerging specification ranges and price indices, its value depends on metal content, impurity profile, moisture, and refinery acceptance, so markets treat it as a quality-differentiated intermediate that is hard to price uniformly (IEA, 2024; Makuza et al., 2021; Yan et al., 2023).

As strategic input, black mass contributes to the secondary supply that the Batteries Regulation and Critical Raw Materials Act frame as reducing third-country dependence, making it an instrument of strategic autonomy (Section 2.2; IEA, 2024).

These three identities do not align. Waste law is organized around risk and control: the Basel Convention defines waste categories and hazardous characteristics and subjects transboundary movements to prior informed consent (Basel Convention, 1989, Annexes I, III, VIII–IX and Art. 6), while the Waste Shipment Regulation adds electronic traceability (European Parliament and Council of the European Union, 2024a). Recycling markets are organized around quality and payable value; critical minerals policy around security of supply. The same material is thus a waste requiring control, a feedstock requiring trade and scale, and a strategic resource requiring support. Classification becomes a governance problem because each regime asks a different question of the same material.

The technical literature adds another reason: black mass is produced through industrial choices. Shredding liberates material cheaply but yields heterogeneous, lower-purity products, while selective disassembly preserves more value at economics that depend on labor, automation, design, and scale (Thompson et al., 2021). Lithium recovery depends on speciation and impurity chemistry (Yan et al., 2023), recovering the other metals requires managed thermal, mechanical, and hydrometallurgical steps (Makuza et al., 2021), and residual electrolyte is hazardous and hard to recover safely (Niu et al., 2023). Value, hazard profile, and tradeability therefore depend on how batteries are collected, sorted, treated, stored, and transported; circular governance depends on the organization of material flows across firms and jurisdictions, not on recycling technology alone.

International waste governance compounds this. Basel and the Waste Shipment Regulation control transboundary movements of the same materials the Batteries Regulation and Critical Raw Materials Act seek to recover, so recovery-oriented materials may remain subject to waste controls under divergent national interpretations (Moisé and Rubínová, 2023). The IEA calls for a harmonized international classification of battery waste and black mass (IEA, 2024), and Zhang et al. (2025) describe black mass as both a valuable resource and a potentially hazardous waste whose classification varies by jurisdiction.

The classification problem is also a measurement problem. Black mass is imperfectly visible in trade statistics because it may be recorded under broader codes for battery scrap, metal-bearing residues, or chemical preparations. Only in March 2025 did the Commission update the European List of Waste to introduce dedicated hazardous-waste codes for waste lithium-based batteries (16 06 07*) and a new chapter (19 14) for the intermediate fractions of their treatment, which its explanatory memorandum identifies as those "commonly referred to as black masses" (Commission Delegated Decision (EU) 2025/934). That these categories had to be created so late is itself evidence: a material of acknowledged strategic and hazardous significance lacked a settled classification throughout the period observed here. The update is not a cause of the trade patterns documented below but confirmation that black mass became a governance object before it was visible in statistical, customs, and waste-classification systems.

Black mass therefore exposes the basic tension in battery circularity policy: the material that makes circularity operational is legally ambiguous, technically heterogeneous, commercially valuable, and internationally mobile. Governing it requires systems for classifying, tracing, transporting, processing, and pricing secondary flows across jurisdictions, not recycling targets alone. It is a test case for whether battery circularity can be governed simultaneously as environmental, industrial, and strategic policy.

2.4 Policy problem: visibility, jurisdiction, and dependence

If black mass is a cross-regime object, the policy problem is not how to recycle it but how to govern a material whose identity differs across waste law, commodity markets, and critical minerals policy. This produces three connected problems: visibility, whether black-mass-related

materials can be identified, measured, and traced; jurisdiction, whose rules attach when the material crosses borders or changes legal status; and dependence, whether circularity reduces strategic vulnerability or relocates it to secondary chokepoints such as refining capacity, the member states through which flows pass, and the bilateral corridors they move through.

Visibility comes first because recycled-content rules, recovery targets, due diligence obligations, and the battery passport all assume that materials remain legible as they move. Black mass is not, because it does not map onto a single statistical, customs, or legal category (Gherasim and Michel, 2024; IEA, 2024). This is a state-capacity problem, not merely an informational one: regulators cannot monitor or strategically manage flows they only partially see. It is now recognized multilaterally, with the United Nations Panel on Critical Energy Transition Minerals calling for a global traceability framework spanning the mineral value chain from mining through recycling (United Nations, 2024).

Jurisdiction follows because circular battery policy is framed through regional objectives (recover materials, retain value, reduce waste exports), while black mass moves through corridors beyond the regulator's reach. Waste law governs shipment legality and consent, battery policy governs producer responsibility and recovery, and critical-minerals policy governs supply security, so "who governs it?" cannot be answered by looking at where the battery was consumed or the recycling target adopted. Jurisdiction depends on how the material is classified, where it is processed, and which legal identity attaches at each border crossing.

Dependence follows because, as Section 2.2 argued, circularity reorganizes rather than eliminates reliance. A region may meet recycling targets while relying on external refiners, retain waste batteries while exporting intermediate fractions, or reduce exposure to mined metals while increasing exposure to a few recycling hubs and corridors. The relevant question is not whether black mass contributes to secondary supply but where control over its transformation is located. Recent research reinforces this point. Wesselkämper et al. (2024) project that China will reach circularity break-even for lithium and nickel more than ten years earlier than Europe and the United States, and Ma et al. (2025) show that efficient recycling depends on standardization and collaboration between recyclers and manufacturers as much as on technology.

This triad positions the paper against prior material-flow and trade research. Sun et al. (2017) showed that lithium resource efficiency cannot be understood through domestic production and consumption alone: lithium commodities move through trade-mediated pathways, and producers lose the ability to trace products after export, so cross-border recycling requires cooperation between trade partners. This paper shifts the object of analysis from upstream lithium flows (mining, chemical production, and manufacture) to black-mass-related trade as a governance problem produced by classification instability, and it moves from the capacity maps and industry interviews on which Gherasim and Michel (2024) relied to the structure of EU export declarations themselves.

Regulatory signaling intensifies all three problems. During 2022-2025, the Batteries Regulation, the Critical Raw Materials Act, and the recast Waste Shipment Regulation signaled tighter expectations around traceability, recovery, shipment control, and supply security. Because most obligations began applying only late in or after the observed trade years, this paper treats these instruments not as causal shocks but as a signaling environment in which recyclers, refiners, and traders could anticipate constraints and opportunities. In annual trade data, such signaling is more likely to appear as changes in routing, concentration, and corridor stability than as compliance effects.

Trade structure is therefore part of the governance problem, and it motivates the empirical strategy. If black mass were governed as a local waste stream, circularity policy would produce transparent, regional, diversified flows, and analysis could focus on collection rates and recovery targets. If instead black-mass-related materials move through concentrated hubs and recurrent bilateral corridors, governance depends on where the material passes as well as on regulatory design: the member states that sort and process it shape access to refining, dominant corridors create vulnerability to disruption, and concentrated supplier-buyer relationships reproduce dependence within nominally circular systems. This is why the paper measures concentration, regionalization, corridor persistence, and bilateral dependence rather than aggregate volumes alone.

3. Methodology

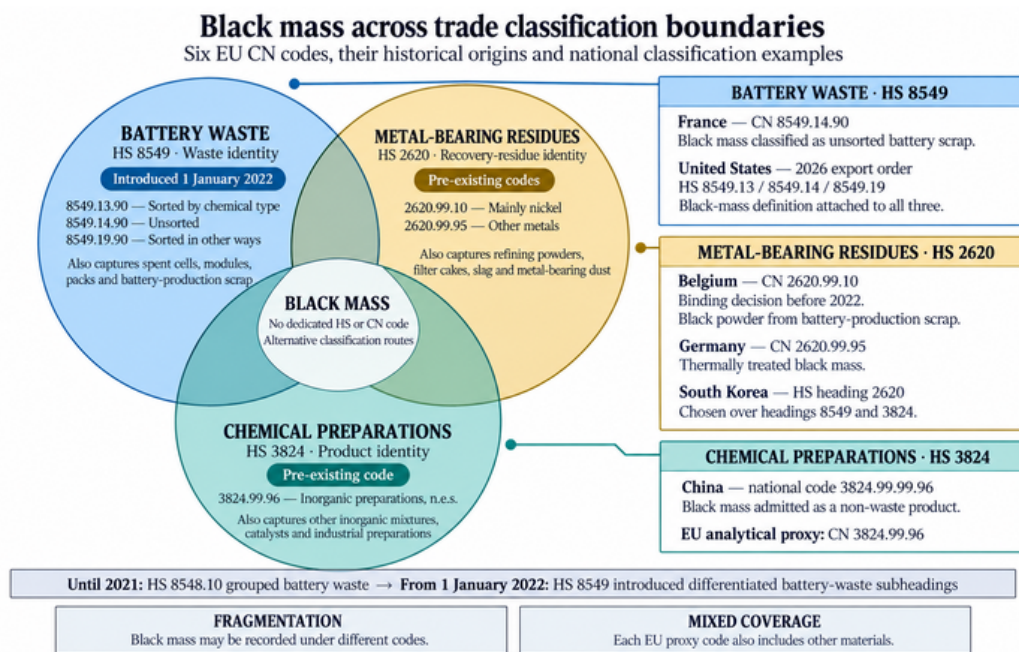
3.1 Products

Black mass has no code of its own in the Harmonized System (HS) or in the EU's Combined Nomenclature (CN). Until 2021, the HS recorded all battery waste under a single subheading, 8548.10, so that spent lead-acid, nickel-cadmium, and lithium-ion batteries were indistinguishable in trade statistics. The 2022 HS edition created heading 8549 for electrical and electronic waste and scrap and divided battery waste into five subheadings: 8549.11 for lead-acid batteries; 8549.12 for other batteries containing lead, cadmium, or mercury; and, for batteries containing none of those metals, 8549.13 sorted by chemical type, 8549.14 unsorted, and 8549.19 sorted in other ways. The CN divides these last three at the eighth digit into whole spent primary cells (.10) and waste and scrap of cells and accumulators (.90). The HS defines electrical and electronic waste as articles rendered unusable for their original purpose. It does not name the powder produced by shredding batteries, and the World Customs Organization has issued no classification opinion on black mass. Whether that powder is battery waste under heading 8549, a metal-bearing residue under heading 2620, or a chemical preparation under heading 3824 – the latter two long-standing HS headings for slag and for miscellaneous chemical preparations – has been left to national customs authorities. Therefore, 2022 is the first year in which lithium battery scrap can be observed in trade data at all.

Customs authorities have placed black mass under all three headings (see Figure 1). Belgium, in a binding decision issued before the 2022 codes existed, classified black powder from

battery production scrap under 2620.99.10, residues containing mainly nickel; Germany classified thermally treated black mass under 2620.99.95, residues containing other metals; and France classified black mass under 8549.14.90, unsorted battery scrap. Korea's customs classification committee weighed all three headings and chose 2620, China admitted black mass as a non-waste product under its national line 3824.99.99.96, chemical preparations, and the United States scoped its 2026 export order to 8549.13, 8549.14, and 8549.19, attaching a definition of black mass to each. The three headings correspond to the three identities described in Section 2.3: 8549 is a waste heading, 2620 is a residue heading for material destined for metal recovery, and 3824 is a product heading. The identity an authority assigns determine the code, and the code decides what trade statistics record. The EU's own machinery for connecting the two has not kept pace: the only EU act that correlates customs codes with waste codes, Commission Implementing Regulation (EU) 2016/1245, still points battery waste to the 8548.10 lines abolished in 2022, so since the new codes began there has been no official bridge between EU customs statistics and EU waste law.

Figure 1. Black mass across trade classification boundaries: six CN proxy codes and national classification decisions



Note: Overlaps are conceptual: they indicate alternative classifications of the same material, not multiple codes assigned to one shipment. Circle areas do not represent trade volumes. Each proxy code also covers materials other than black mass. Source: Authors' compilation; see Table 1.

The analysis therefore follows six CN codes (Table 1). Three are the battery scrap codes created in 2022: 8549.13.90, 8549.14.90, and 8549.19.90, the waste-and-scrap lines under the three subheadings for batteries without lead, cadmium, or mercury. The parallel .10 lines are excluded because they carry whole spent primary cells, which are non-rechargeable and therefore not lithium-ion, and subheadings 8549.11 and 8549.12 because lead-acid and cadmium or

mercury chemistries are not lithium-ion. Three are older codes into which customs authorities have placed black mass: 2620.99.10 and 2620.99.95, the two refinery residue codes, and 3824.99.96, the chemical mixture code corresponding to the Chinese national line. Table 1 gives each code's legal text, the variety of goods that physically move under it, and whether that material is hazardous waste under EU law, a designation that applies to waste lithium-based batteries and black mass only from November 2026.

Table 1. Black mass-relevant EU Combined Nomenclature codes, 8-digit

CN code	CN text	Examples of what physically travels	Hazardous waste under EU law?	Ex. customs authorities that classify black mass here (code)
8549.13.90	Battery waste and scrap, sorted by chemical type, without lead, cadmium or mercury	Black mass in one-tonne woven bags, wrapped on pallets; pallets of drums and boxes of sorted alkaline and zinc-carbon cells from a sorting plant; whole, spent lithium-ion modules and cells; electrode cut-offs and reject cells from battery plants.	Yes	United States (HS 8549.13)
8549.14.90	Battery waste and scrap, unsorted, without lead, cadmium or mercury	Black mass in one-tonne woven bags, wrapped on pallets; pallets of drums and boxes of mixed consumer batteries from collection points, going to a sorting plant.	Yes	France; United States (HS 8549.14)
8549.19.90	Battery waste and scrap, sorted but not by chemical type, without lead, cadmium or mercury	Black mass in one-tonne woven bags, wrapped on pallets; spent whole EV packs and modules on pallets.	Yes	United States (HS 8549.19)
2620.99.10	Slag, ash and residues containing mainly nickel	Nickel-rich black mass from battery-plant scrap in one-tonne woven bags, wrapped on pallets; drums and bags of powders and filter cakes from nickel refining and hydrometallurgy.	In part	Belgium; Korea (HS 2620.99)
2620.99.95	Slag, ash and residues containing other metals or metal compounds	Thermally treated black mass in one-tonne woven bags, wrapped on pallets; slag and flue dust in bulk; drums and lined bags of sludge or dried cake from copper refineries carrying gold, silver, platinum-group metals and selenium.	In part	Germany; Korea (HS 2620.99)
3824.99.96	Inorganic chemical products and preparations, n.e.s.	Black mass in one-tonne woven bags, wrapped on pallets; drums, intermediate bulk containers and bags of inorganic mixtures without their own line: catalysts, water-treatment blends, cement additives, desulphurisation reagents, cathode precursor powders.	In part	China (national code 3824.99.99.96)

Note: "Hazardous waste under EU law?" refers to the European List of Waste (Commission Decision 2000/532/EC, amended by Delegated Decision (EU) 2025/934). Waste lithium-based batteries and black mass are hazardous from November 2026. Law applies "in part" when the different products traded under the CN code have mixed status.

The six codes differ in how closely they track black mass. The three battery scrap codes were created for battery waste and are closest to the object of interest, though they carry other material too. The three older codes cover heterogeneous materials on a far larger scale, which introduces considerable noise. This is a feature of the governance problem rather than a defect of the data.

3.2 Data

The analysis is restricted to EU-origin exports: exporters are the 27 EU member states and destinations are all partner countries, inside and outside the EU. The data are annual export declarations from Eurostat's Comext database at the 8-digit CN level for 2022 through 2025, the four years for which the dedicated battery scrap codes exist, with value in euros and quantity in metric tons. Exports outside the EU come from customs declarations; trade between member states is recorded through Intrastat, the survey system that replaced customs declarations inside the single market. Member states must cover at least 95% of the value of their dispatches to other member states, and traders below national thresholds are exempt (Regulation (EU) 2019/2152, Article 12(2)), so intra-EU flows are lower bounds and the smallest shipments are missing. The data describe what EU member states export, not the global black mass market.

Comext records the code, value, quantity, exporting member state, and destination of each flow. It does not record whether a shipment was declared as waste, the battery chemistry, or the composition of the material. The analysis uses exporters' declarations throughout rather than mirror data from importers, so every flow is measured once, by the member state that exported it. This choice matters for materials without agreed classifications. Reconciled bilateral trade databases such as BACI (Gaulier and Zignago, 2010) average exporter and importer reports, and where the importer records the same goods under a different heading, the reconciled flow shrinks toward zero. For example, Spain's exports to South Korea under HS 2620.99 are €143 million, €181 million, and €144 million in 2022, 2023, and 2024 in Comext; in BACI, the same flow is \$145 million, \$71 million, and zero. A flow that the two ends of the corridor classify differently disappears from reconciled data and survives only in the declarations of one side: the visibility problem reaches into the statistical sources themselves.

Our empirical design reflects the paper's policy question: how EU-origin black-mass-related material flows are organized during a period of battery-circularity regulatory signaling. Specifically, the period includes several relevant regulatory actions: the 2023 adoption and entry into force of the EU Batteries Regulation; the 2024 recast EU Waste Shipment Regulation; the 2024 Critical Raw Materials Act; and continuing Basel Convention technical work on waste batteries and black mass in which the EU has been closely involved (Appendix Table A1). It ends with the 2025 update to the European List of Waste, which from 9 November 2026 classifies waste lithium batteries and black mass as hazardous (Commission Decision 2000/532/EC as amended by Delegated Decision (EU) 2025/934). Table 1 records which codes this reclassification will

affect, and the contrast between the three battery scrap codes and the three older codes runs through Section 4.

The final sample contains 9,972 exporter-importer-product-year observations, spanning 27 EU exporters, 223 destination economies, and 2,808 unique bilateral exporter-importer corridors (2022-2025).

3.3 Measures

Let k index CN codes, e index EU-27 exporters, and i index importers. Let v_{eik} be the export value in euros from e to i in code k , and q_{eik} the quantity in metric tons, each summed over 2022-2025. All measures are computed on these sums unless a year is stated. Write $V_k = \sum_e \sum_i v_{eik}$ for the code's total export value.

Unit value of code k :

$$u_k = \frac{\sum_e \sum_i v_{eik}}{\sum_e \sum_i q_{eik}}.$$

The unit value of a single flow is $u_{eikt} = \frac{v_{eikt}}{q_{eikt}}$, computed at the exporter-destination-year level. The share of code k 's quantity in unit-value band b is the quantity of the flows whose unit value lies in the band over the code's quantity:

$$w_{bk} = \frac{\sum_{(e,i,t): u_{eikt} \in b} q_{eikt}}{\sum_e \sum_i q_{eik}}.$$

Importer i 's share of code k , the concentration index, and the effective number of importers:

$$s_{ik} = \frac{\sum_e v_{eik}}{V_k}, \quad \text{HHI}_k = \sum_i s_{ik}^2, \quad N_k^{\text{eff}} = \frac{1}{\text{HHI}_k}.$$

The HHI approaches its lower bound $\frac{1}{N_k}$, where N_k is the number of importers with positive value, when trade is spread evenly across importers, and equals 1 if all EU-27 exports of code k go to one importer. We compute the same three measures for exporters e .

Share of code k 's value going to destination region g , where the sum runs over importers in g :

$$r_{gk} = \frac{\sum_{i \in g} \sum_e v_{eik}}{V_k}.$$

A corridor is an exporter-importer pair (e, i) . Its share of code k , and its dependence measured at each end:

$$c_{eik} = \frac{v_{eik}}{V_k}, \quad D_{eik}^{\text{exp}} = \frac{v_{eik}}{\sum_i v_{eik}}, \quad D_{eik}^{\text{imp}} = \frac{v_{eik}}{\sum_e v_{eik}}.$$

A corridor is active in year t if $v_{eikt} > 0$.

Intra-EU exports of member state m in code k are $\sum_{i \in \text{EU}-27} v_{mik}$; its intra-EU imports are $\sum_{e \in \text{EU}-27} v_{emk}$.

4. Results

The results show that EU-origin black-mass-related trade is not organized as a single, diffuse circular market. Observable patterns connect back to the three governance problems discussed in Section 2.4: visibility, jurisdiction, and dependence.

4.1 Visibility: Not a single market

Table 2 reports the value, quantity, and unit value of EU-27 exports under the six CN codes, summed over 2022-2025. The codes differ by orders of magnitude: the three battery scrap codes carried €51 million to €79 million each, the two refinery residue codes carried €40 million and €2.55 billion, and the chemical mixture code €29.6 billion. Unit values range from €513 to €3,012 per ton. Black mass is not one product in trade data. It is a set of codes of very different sizes and content. The three battery scrap codes are the ones closest to black mass: under them move black mass in one-ton bags, whole spent lithium-ion cells and modules, and sorted alkaline and zinc-carbon cells from sorting plants. Under the two refinery residue codes move slag, flue dust, filter cakes, and the sludges from copper and nickel refining that carry gold, silver, and platinum-group metals, alongside black mass. The chemical mixture code is the residual line for inorganic chemical preparations with no line of their own: under it move catalysts, water-treatment blends, cement additives, and cathode precursor powders, alongside black mass. It is the largest code in Table 2 by a factor of more than ten.

The codes are also heterogeneous inside. A CN code is a category, not a product, and shipments under the same code sell at very different prices per ton.

The last three columns of Table 2 divide each code's tonnage into three unit-value bands: below €1,000 per ton, €1,000 to €10,000 per ton, and above €10,000 per ton. For reference, black mass sold for about €3,200 to €4,100 per ton in 2024 (Fastmarkets, 2024). Heterogeneity is present even within the three dedicated battery scrap codes. Under the sorted-by-chemistry code, 65% of the tonnage is in the lowest band and 35% in the middle band. Under the sorted-but-not-by-chemistry code, the tonnage is split almost evenly between the lowest and middle bands. Only the unsorted code has most of its tonnage in a single band, the middle one. The three legacy codes are heterogeneous in the same way. Under the miscellaneous metals residue code, 90% of the tonnage is in the lowest band, while 91% of the value is in the highest band. Under the nickel residue code, four fifths of the tonnage is in the lowest band. Under the chemical mixture code, tonnage falls in

all three bands. Each of the six codes carries several goods at once, and black mass is one of them. This is the visibility problem in the trade data themselves.

Table 2: Heterogeneity among black mass products (2022-2025)

CN code	Short label	Value (€ million)	Quantity (tons)	Unit value (€ per ton)	Share of quantity by unit value		
					Under €1,000 per ton	€1,000 to €10,000 per ton	Over €10,000 per ton
8549.13.90	Battery scrap, sorted by chemistry	55.3	53,499	1,034	65%	35%	0%
8549.14.90	Battery scrap, unsorted	51.4	29,847	1,720	14%	86%	0%
8549.19.90	Battery scrap, sorted but not by chemistry	78.6	40,060	1,961	46%	52%	2%
2620.99.10	Refinery residues, mainly nickel	40.0	78,016	513	81%	19%	0%
2620.99.95	Refinery residues, misc. metals	2,550.7	846,731	3,012	90%	9%	1%
3824.99.96	Mineral and metal chemical mixtures, misc.	29,566.7	11,319,798	2,612	35%	63%	2%

Note: Summed EU-27 export declarations, 2022-2025 (Eurostat Comext). Quantity in metric tons. Unit-value bands are computed on each exporter-destination-year flow as declared value over declared quantity; a flow that mixes goods takes a blended unit value, so the bands understate how mixed a code is. Value shares by band, cited in the text, are computed the same way. The reference price for black mass converts Fastmarkets' 2024 dollar assessments at the 2024 average exchange rate.

4.2 Jurisdiction: Reach and limits of EU rules

Table 3 reports the share of each code's export value by destination region, with the EU-27 shown separately.¹ Unsorted battery scrap stays almost entirely inside the EU-27, at 93%. Battery scrap sorted by chemistry is split: 56% to the EU-27 and 40% to Eastern and Southeastern Asia. Battery scrap sorted but not by chemistry leaves: 31% to the EU-27 and 60% to Eastern and Southeastern Asia. The nickel residue code stays in the EU-27 at 98%; the miscellaneous metals residue code goes 42% to the EU-27 and 51% to Eastern and Southeastern Asia. Destinations outside these regions take under 4% of any battery scrap or refinery residue code. The chemical mixture code stands apart. It sends 18.5% of its value to regions that take almost none of any other code, and 17.2% to non-EU Europe and Northern America, more than double any other code's

¹ Using the UN M49 geographic regions, defined as Sub-Saharan Africa; Northern Africa and Western Asia; Central and Southern Asia; Eastern and Southeastern Asia; Latin America and the Caribbean; Oceania; and Europe and North America.

(<https://unstats.un.org/unsd/methodology/m49/>)

share there. This is consistent with the code carrying mainly other chemical preparations and only a small share of black mass. Of the €82 million of battery scrap exported outside the EU over the period, €43 million went to countries outside the OECD, €39 million of it to Singapore. The two sorted battery scrap codes are the ones that leave the EU; the unsorted code stays.

Table 3: Importer destination regions (2022-2025)

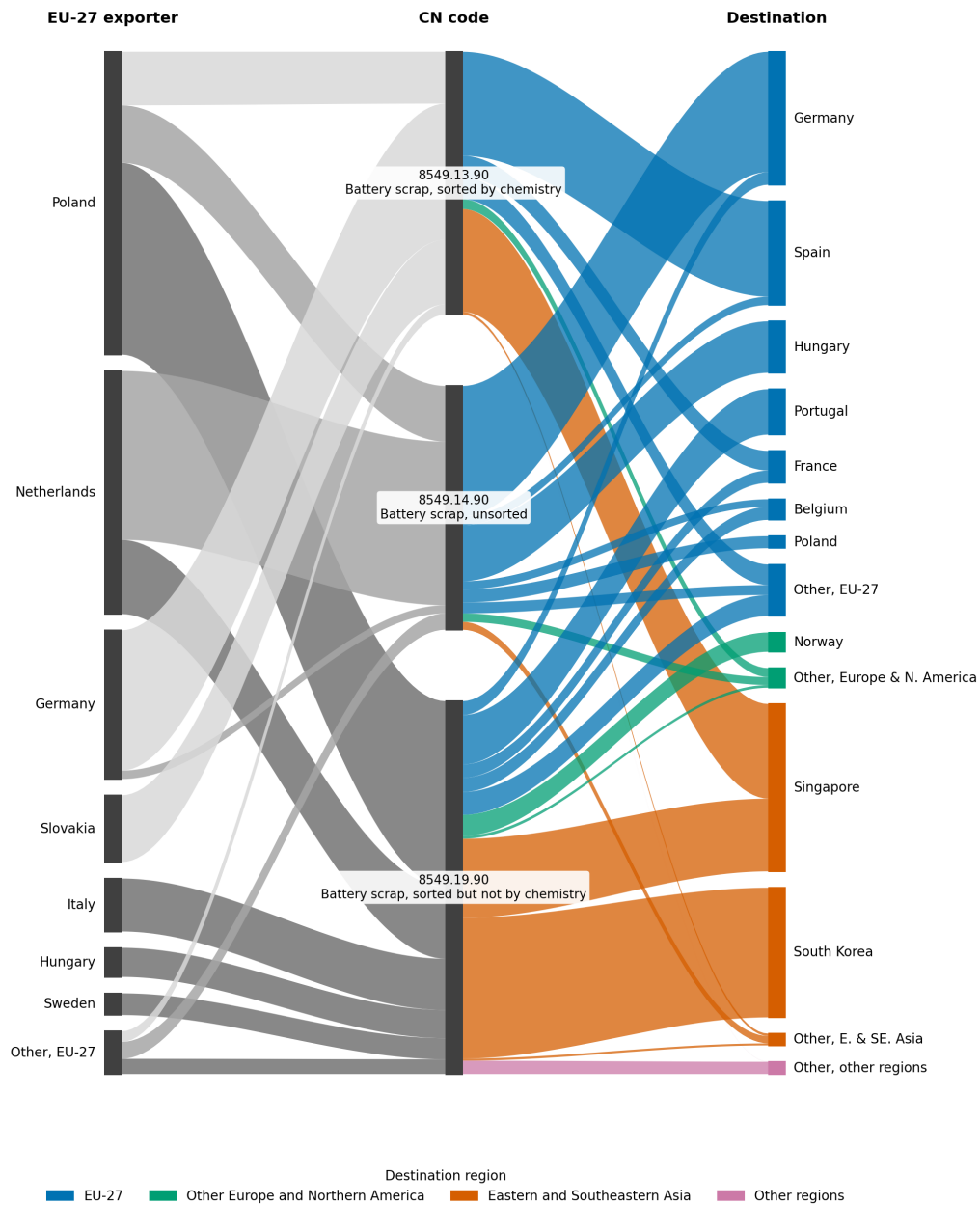
CN code	Short label	EU-27	Other Europe and Northern America	Eastern and Southeastern Asia	Other regions
8549.13.90	Battery scrap, sorted by chemistry	56.1%	3.7%	40.1%	0.0%
8549.14.90	Battery scrap, unsorted	93.3%	3.5%	3.2%	0.0%
8549.19.90	Battery scrap, sorted but not by chemistry	30.5%	6.4%	59.5%	3.6%
2620.99.10	Refinery residues, mainly nickel	98.0%	1.0%	1.0%	0.0%
2620.99.95	Refinery residues, misc. metals	41.7%	7.2%	51.0%	0.2%
3824.99.96	Mineral and metal chemical mixtures, misc.	56.0%	17.2%	8.4%	18.5%

Note: Shares of EU-27 export value summed over 2022-2025 (Eurostat Comext), by destination region. Regions are United Nations M49 groupings, with the Europe and Northern America group split into the EU-27 and the rest. Other regions: Central and Southern Asia, Northern Africa and Western Asia, Latin America and the Caribbean, Sub-Saharan Africa, Oceania, and destinations not specified.

Figure 2 shows the same flows for the three battery scrap codes as bands from exporting member state, through CN code, to destination, with band width proportional to value. Three exporters account for most of the material: Germany, which mainly ships scrap sorted by chemistry; the Netherlands, which mainly ships unsorted scrap; and Poland, which ships under all three codes. The code a shipment carries is largely a property of the exporter rather than of the destination. The main intra-EU destinations are Germany, Spain, Hungary, and Portugal; the Asian destinations are Singapore and South Korea.

Those two are where the refining of black mass into recovered metals is concentrated. South Korea's post-treatment capacity grew over 2022-2024 but not as fast as inflows, and its OECD membership lets it receive waste shipments from the EU under the simplified procedure (Gherasim and Michel, 2024). Singapore hosts SK TES, a battery recycler owned since 2022 by the Korean SK group (Korea Herald, 2022).

Figure 2. EU-27 exports of battery scrap by exporter, CN code and destination (2022-2025)



Note: EU-27 export value summed over 2022-2025 (Eurostat Comext); band width is proportional to value. Exporters and destinations below 3% of a CN code's value are grouped as "other."

4.3 Dependence: Corridor concentration

Tables 4 and 5 report, for each code, the number of exporting member states and of destination countries, the Herfindahl-Hirschman Index (HHI) of their value shares, the effective

number of partners, and the three largest partners with their shares. On the exporter side, the three battery scrap codes and the two refinery residue codes have effective numbers of two to three exporters, with HHI values between 0.31 and 0.51; the chemical mixture code has 5.8 effective exporters and an HHI of 0.17. On the importer side, the battery scrap codes have three to five effective destinations, the nickel residue code two, the miscellaneous metals residue code four, and the chemical mixture code 23. In every battery scrap and refinery residue code, the largest partner on one side or the other holds around half or more of the value: Germany 52% of sorted-by-chemistry exports, the Netherlands 67% of unsorted exports, Germany 54% of unsorted imports, France 65% of imports under the nickel residue code. Concentration of this degree on both sides, high by HHI standards, with a single dominant partner on at least one side in every code except the chemical mixture code, is the first form the dependence problem takes.

Table 4: Exporter-side heterogeneity (2022-2025)

CN code	Short label	Exporters (total count)	Exporter concentration (HHI)	Effective no. of exporters	Top three (% shares)
8549.13.90	Battery scrap, sorted by chemistry	15	0.368	2.7	Germany (51.7%), Slovakia (24.6%), Poland (19.7%)
8549.14.90	Battery scrap, unsorted	16	0.505	2.0	Netherlands (67.1%), Poland (22.8%), Germany (3.3%)
8549.19.90	Battery scrap, sorted but not by chemistry	21	0.314	3.2	Poland (49.8%), Netherlands (19.3%), Italy (13.8%)
2620.99.10	Refinery residues, mainly nickel	16	0.493	2.0	Netherlands (67.8%), Belgium (15.3%), Greece (8.5%)
2620.99.95	Refinery residues, misc. metals	21	0.417	2.4	Spain (56.2%), Finland (31.4%), Germany (2.7%)
3824.99.96	Mineral and metal chemical mixtures, misc.	26	0.174	5.8	Germany (28.0%), Ireland (25.0%), France (12.1%)

Note: Shares of EU-27 export value summed over 2022-2025 (Eurostat Comext), by exporting member state.

Table 5: Importer-side heterogeneity (2022-2025)

CN code	Short label	Importers (total count)	Importer concentration (HHI)	Effective no. of importers	Top three (% shares)
8549.13.90	Battery scrap, sorted by chemistry	37	0.318	3.1	Spain (39.5%), Singapore (39.1%), France (8.1%)
8549.14.90	Battery scrap, unsorted	35	0.346	2.9	Germany (53.5%), Hungary (22.9%), Poland (5.3%)
8549.19.90	Battery scrap, sorted but not by chemistry	65	0.214	4.7	South Korea (37.8%), Singapore (21.2%), Portugal (13.2%)
2620.99.10	Refinery residues, mainly nickel	38	0.457	2.2	France (65.1%), Netherlands (15.4%), Germany (9.3%)
2620.99.95	Refinery residues, misc. metals	87	0.225	4.4	Sweden (30.0%), Japan (27.5%), South Korea (23.5%)
3824.99.96	Mineral and metal chemical mixtures, misc.	223	0.043	23.0	Germany (10.3%), France (7.9%), United States (6.2%)

Note: Shares of EU-27 export value summed over 2022-2025 (Eurostat Comext), by destination country.

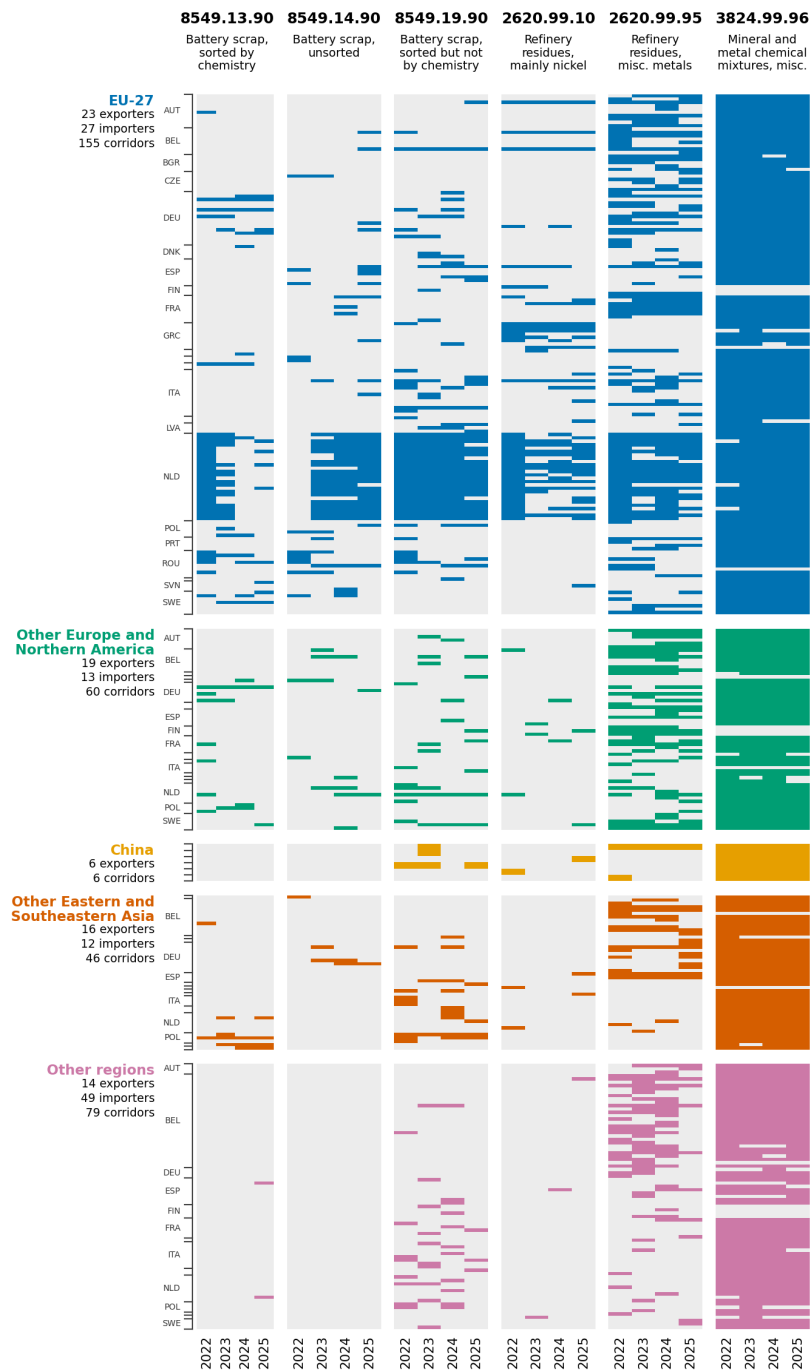
Table 6 reports the leading exporter-importer corridor in each code, its value, and how much of the trade flowed through it: as a share of the code, of the exporter's total, and of the importer's total. In the battery scrap codes the leading corridors are Germany to Spain, €21 million and 38% of its code; the Netherlands to Germany, €27 million and 53%; and Poland to South Korea, €22 million and 28%. In each, the importer takes 74% to 99% of its supply from that one exporter. In the refinery residue codes the corridors are larger and more complete, and in one case far larger: the Netherlands to France carries 64% of the nickel residue code, and Finland to Sweden carries €764 million under the miscellaneous metals residue code, effectively all of Sweden's imports and 95% of Finland's exports. In the chemical mixture code no corridor exceeds 5% of the code. Dependence in the black-mass-related codes is bilateral: it takes the form of one exporting country and one destination country that each account for most of the other's trade.

Table 6: Dominant corridors by CN code (2022-2025)

CN code	Short label	Leading corridor	Value (€ million)	What share of trade flowed through this corridor?		
				Overall	For the exporter	For the importer
8549.13.90	Battery scrap, sorted by chemistry	Germany to Spain	20.9	38%	73%	95%
8549.14.90	Battery scrap, unsorted	Netherlands to Germany	27.2	53%	79%	99%
8549.19.90	Battery scrap, sorted but not by chemistry	Poland to South Korea	22.0	28%	56%	74%
2620.99.10	Refinery residues, mainly nickel	Netherlands to France	25.5	64%	94%	98%
2620.99.95	Refinery residues, misc. metals	Finland to Sweden	763.9	30%	95%	100%
3824.99.96	Mineral and metal chemical mixtures, misc.	Ireland to Mexico	1,400.3	5%	19%	80%

Figure 3 shows, for every exporter-importer corridor that traded under at least one of the five battery scrap and refinery residue codes, which of the four years it traded in, with one panel per code and rows grouped by destination region. The chemical mixture code is shown for comparison in the sixth panel, restricted to those same corridors. In the chemical mixture code most corridors are filled in all four years. In the two refinery residue codes about a quarter of corridors traded in every year. In the three battery scrap codes 6%, 0%, and 20% of corridors did, and the largest corridors of 2022 are not those of 2025: Poland to Hungary carried 92% of unsorted scrap in 2022 and nothing after 2022, while the Netherlands to Germany carried nothing in 2022 and two thirds or more of the code from 2023. The corridors on which the battery scrap codes depend are one to three years old.

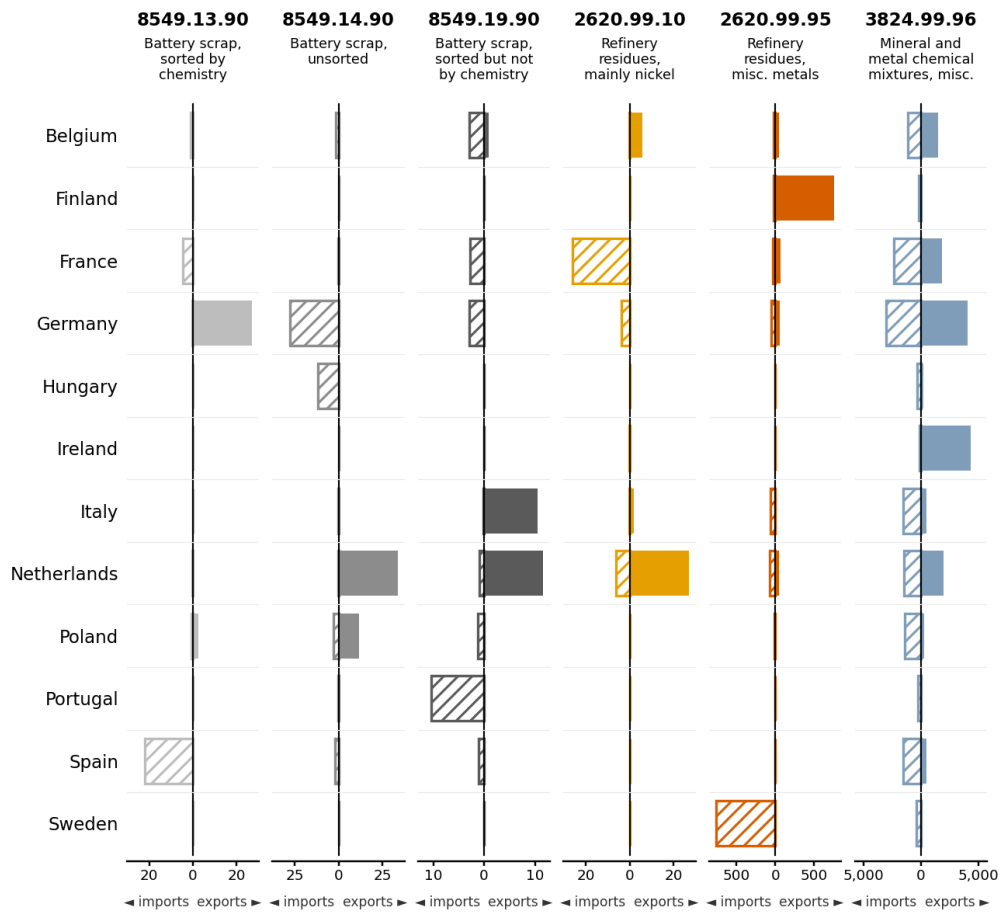
Figure 3. Exporter-importer corridors trading under each CN code, by year (2022-2025)



Note: Each row is an EU-27 exporter-importer corridor. A colored cell indicates a positive export value. CN 3824.99.96, which covers miscellaneous chemical mixtures, is traded in an additional 2,462 corridors in the period; these are excluded for presentation purposes. Source: Eurostat Comext.

Figure 4 shows, for the member states with at least 10% of intra-EU trade in any code, what each exported to and imported from other member states under each code. Most member states appear on one side only. The Netherlands, Poland, and Italy export battery scrap and import little; Spain, Hungary, Portugal, and France import it and export little. Germany is the exception. Its inbound and outbound streams are different goods. Over 2022-2025 Germany imported €27.2 million of unsorted battery scrap, 18,400 tons, from the Netherlands at about €1,500 per ton. Its exports under the sorted-by-chemistry code, €28.6 million and 38,100 tons, split by price into two streams. Spain took €20.9 million on 4,100 tons, about €5,100 per ton, and France €3.7 million on 550 tons, about €6,700 per ton; these are lithium-ion prices. Bulgaria took 19,300 tons for €0.5 million, about €26 per ton, and Poland 11,900 tons for €0.8 million, about €67 per ton; these are the prices of sorted alkaline and zinc-carbon cells going to zinc smelters. By weight, four fifths of what Germany exports under the sorted-by-chemistry code is the low-value stream; by value, nine tenths is the lithium-ion stream to Spain and France. The same CN code carries both. Hungary imported €12 million of unsorted scrap from Poland in 2022 alone. Within the EU, the processing of battery scrap runs through one member state, and the corridor into it is the one that formed in 2023.

Figure 4. Intra-EU trade in black mass-related CN codes, selected member states (2022-2025)



Note: € million, summed over 2022-2025. Member states shown account for at least 10% of intra-EU trade in at least one code. Source: Eurostat Comext.

5. Discussion of empirical findings

Together, the results show that EU-origin black-mass-related trade is visible only through fragmented product categories, organized through a handful of dominant corridors, and dependent on a small set of member states and bilateral relationships. The visibility problem is shown by the six-code design itself, not by any single measure. The strongest concentration and dependence patterns appear not in the chemical mixture code, but in the battery scrap and refinery residue codes. Battery circularity therefore does not eliminate trade dependence; it reorganizes it across sorting and processing sites, trade corridors, and classification boundaries. The corridors that carry it are also new: few battery scrap corridors traded in all four years (Figure 3), so the concentration rests on relationships that are one to three years old and have not yet shown they will last.

The heterogeneity in Table 2 is itself the paper's first governance finding. Visibility, unlike concentration or dependence, cannot be captured by a single index, because the absence of a dedicated black-mass code is precisely what makes the material hard to govern. The fact that black-mass-related trade can only be approximated through six codes, each carrying black mass alongside goods worth prices per ton differ a hundredfold is a finding rather than a limitation: a regulator attempting to monitor these flows faces the same classification ambiguity this analysis confronts. The IEA has itself called for a harmonized international classification of black mass to reduce regulatory uncertainty (IEA, 2024); indicated that the gap is recognized at the level of minerals governance, not merely an artefact of this paper's data choices.

The jurisdiction problem has a concrete case in the Singapore corridor. Over 2022-2025, €22 million of battery scrap sorted by chemistry (CN 8549.13.90) went from the EU to Singapore, 89% of that code's exports outside the EU, at unit values of €1,200 to €7,100 per ton, the range in which black mass trades. Singapore is not a member of the OECD. During the study period, these shipments were lawful under EU waste law. Sorted waste batteries conforming to a specification are on the Basel Convention's list of non-hazardous wastes, and black mass had no entry of its own in the European List of Waste and was not classified as hazardous, so neither fell under the prohibition on exporting hazardous waste for recovery to countries outside the OECD (Regulation (EC) No 1013/2006, Article 36(1)).

From 9 November 2026, black mass from lithium batteries is listed as hazardous waste under entry 19 14 02* (Commission Delegated Decision (EU) 2025/934). From that date, the recast Waste Shipment Regulation prohibits the export of wastes listed as hazardous in the European List of Waste for recovery in countries to which the OECD Decision does not apply (Regulation (EU) 2024/1157, Article 39(1)(b)). Material declared as waste can then no longer be shipped to Singapore for recovery. The trade can continue on one condition: that the material is declared as a product rather than as waste, and accepted as such by the competent authority, which places it outside the regulation. The EU's own reclassification therefore does not close the corridor. It moves the decision about whether the corridor stays open from the waste regulator to the exporter's classification of the shipment, and it does so for the corridor carrying the largest share of lithium-relevant material out of the EU. Customs data do not record whether a shipment was declared as waste or as product, so the extent to which exporters have already taken this route cannot be observed; the price dispersion in Section 4.1 suggest that the market pays more for material with product status.

The policy implication is important. EU battery circularity may strengthen regional control over collection and early-stage processing, but the refining stage stays outside the EU: EU-origin black-mass-related flows combine regional waste handling with extra-European refining. Circularity, in this sense, is not the opposite of trade dependence; it is a reconfiguration of where dependence appears in the material chain.

The dependence problem can be measured against the EU's own yardstick. The Critical Raw Materials Act sets a benchmark that by 2030 no more than 65% of the EU's annual consumption of any strategic raw material, at any stage of processing, should come from a single third country (Regulation (EU) 2024/1252). The benchmark is written for imports. Applied to exports of the material the EU wants to recover, the same threshold is exceeded in the sorted by chemistry code. Of battery scrap sorted by chemistry (CN 8549.13.90) exported outside the EU over 2022-2025, 89% went to Singapore. Of battery scrap sorted but not by chemistry (CN 8549.19.90) exported outside the EU, 54% went to South Korea. Across the three battery scrap codes together (CN 8549.13.90, 8549.14.90, and 8549.19.90), Singapore took 47% and South Korea 37% of exports outside the EU, so two countries took 84%. The comparison is illustrative rather than legal, since the Act governs where the EU buys, not where it sells. It shows that the concentration the Act treats as a supply risk on the import side already exists on the export side, in the intermediate material that the Act's recycling benchmark depends.

These findings show that battery circularity does not simply replace primary-material dependence with open and diversified secondary-material markets. In several codes, circular and black-mass-related flows appear to reorganize dependence around specific bilateral relationships. From a governance perspective, this creates vulnerabilities around supplier concentration, corridor instability and disruption, regulatory changes, and the location of downstream processing capacity. The dependence problem is therefore not only whether Europe can recover more material, but whether recovered and waste-derived materials move through resilient and diversified channels.

Section 2.4 anticipated that regulatory signaling over 2022-2025 would show up in trade data as changes in routing, concentration, and corridor stability rather than as compliance effects. At the level of codes, the data show no trend. The share of battery scrap export value that stayed inside the EU was 59% in 2022, 53% in 2023, 47% in 2024, and 65% in 2025 (CN 8549.13.90, 8549.14.90, and 8549.19.90 together). The three codes move in different directions: the unsorted code stays above 87% intra-EU in every year, the sorted-by-chemistry code falls from 91% to 30% and returns to 69%, and the sorted-but-not-by-chemistry code stays between 26% and 42%. Concentration among exporters and among importers also varies from year to year without a direction.

At the level of corridors, the data show a great deal of change. As Figure 3 shows, the corridors that carried the battery scrap codes in 2022 are largely not those that carried them in 2025, while the corridors of the refinery residue and chemical mixture codes are the same throughout. Poland to Hungary carried 92% of the unsorted code in 2022 and nothing after; the Netherlands to Germany carried nothing in 2022 and two thirds or more of the code from 2023. With four annual observations and codes that begin in 2022, these changes cannot be attributed to any regulation. They are consistent with a market in which trading relationships were being formed. The dedicated codes date from 2022, so the trade they record has been visible for exactly as long as it has existed; the corridors that policy now addresses are the same age as the data.

6. Conclusion

This paper follows the first four years in which lithium battery scrap can be observed in trade data. It uses EU export declarations under six Combined Nomenclature (CN) codes and measures concentration, corridor dependence, corridor persistence, and regionalization. It examines whether EU battery circularity policy has produced a regional loop. In the study period, we find that the loop is partial. Collection, sorting, and early processing take place largely inside the EU, and much of the refining of sorted material takes place outside it. Black mass is visible only through fragmented and internally heterogeneous codes. Sorted material leaves the EU for refining destinations that the EU's waste rules reach only in part. EU exports depend on a few young and fluctuating corridors.

The three governance problems we examine theoretically each have empirical forms, several of which are worthy of additional attention. On visibility, black mass is not observable as a good in trade statistics and even with improved measurement remains a substance traded under a variety of different codes. On jurisdiction, the largest corridor carrying lithium-relevant material out of the EU is to Singapore, a country outside the OECD. From November 2026, EU waste law prohibits exporting black mass, once listed as hazardous waste, for recovery to non-OECD countries, but does not affect trade in black mass classified as a non-waste product. On dependence, Singapore takes 89% of sorted-by-chemistry battery scrap exported outside the EU, a concentration above the single-country threshold the Critical Raw Materials Act sets for imports. Across the three governance problems, the instrument that decides whether EU rules reach a shipment is the classification the exporter gives it, not the regulation itself. Like other trading states, the EU and its member states regulate trade through classification, and in the context of black mass classification is a complex and moving target.

The global production networks literature describes the state as an actor that can facilitate, prolong, or end a region's coupling to a network. For black mass, the EU's instruments are classifications and targets: customs codes, waste codes, and recovery and recycled-content targets. These instruments act on declarations about the material rather than on the material itself, and the declarations are made by exporters and accepted or contested by authorities. The trading relationships documented here are observable only as far back as the codes that record them. The reclassification of black mass as hazardous waste in November 2026 is the first EU measure that applies to those relationships directly. Future trade data will show whether black mass stays in the EU, leaves as a product, or leaves through countries within the OECD framework. On the evidence of 2022-2025, that outcome depends on classification at least as much as on recycling capacity.

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Appendix.

Table A1. Regulatory Developments 2022-2024

Regulatory development	Brief description	Relevance to black mass	Key dates	Future / phased requirements
EU Batteries Regulation, Regulation (EU) 2023/1542	Establishes requirements on battery sustainability, safety, labelling, marking, information, extended producer responsibility, collection, treatment, reporting, due diligence, and recycled-content requirements.	Relevant because black mass is generated from waste-battery treatment and recycling. The regulation strengthens collection obligations, recycling-efficiency and material-recovery targets, recycled-content reporting, and recovery of cobalt, lithium, nickel, copper, and lead.	Adopted on 12 July 2023; published in the Official Journal on 28 July 2023; entered into force on 17 August 2023; generally applicable from 18 February 2024.	Waste-battery management rules apply from 18 August 2025. Material-recovery targets must be achieved by 31 December 2027 and increase by 31 December 2031. Minimum recycled-content requirements begin on 18 August 2031 and increase from 18 August 2036. Battery due-diligence obligations were originally scheduled to apply from 18 August 2025, but were postponed to 18 August 2027 by Regulation (EU) 2025/1561.
Recast EU Waste Shipment Regulation, 2024/1157	Recasts and updates EU controls on shipments of waste within the EU and between the EU and third countries, including procedures, enforcement, traceability, electronic information exchange, and restrictions on certain exports.	Relevant because black mass may be classified or shipped as waste, residue, battery scrap, or a waste-derived intermediate material. The regulation affects the legal and administrative conditions under which such material can move across borders, especially where its waste/non-waste status, hazard classification, or recovery destination is contested.	Adopted on 11 April 2024; published on 30 April 2024; entered into force on 20 May 2024.	Main application begins on 21 May 2026, with selected provisions applying earlier or later. Accordingly, 2024 is best treated as an entry-into-force and adaptation period rather than a full implementation year.
Critical Raw Materials Act, Regulation (EU) 2024/1252	Creates an EU framework to secure resilient and sustainable supply of critical and strategic	Relevant because black mass contains or is linked to recoverable	Adopted 11 April 2024 ; published 3 May 2024 ; entered into force 23 May 2024 .	Sets 2030 EU capacity benchmarks, including recycling capacity able to produce at least 25% of annual EU consumption

	raw materials, including EU extraction, processing and recycling capacity, strategic projects, supply-risk monitoring, and import diversification.	critical materials such as lithium, cobalt, nickel, and manganese-bearing streams. It reinforces the strategic value of battery recycling and secondary raw-material recovery.		of strategic raw materials and to recycle increasing amounts of each strategic raw material from waste. Additional waste-stream-based recycling-capacity benchmarks are to be developed through delegated acts.
Basel Convention - technical work on waste batteries	Ongoing international work to update technical guidelines for environmentally sound management of waste lead-acid batteries and to develop technical guidelines for waste batteries other than lead-acid batteries.	Relevant because black mass arise from battery-waste treatment and may be treated as waste, residue, recycled material, hazardous material, or secondary raw-material input. Basel work matters because the Convention controls transboundary movements of hazardous and other wastes and requires environmentally sound management.	COP-15 initiated the work in 2022 ; COP-16 extended the mandate in 2023 ; technical drafting and comments continued through 2024 and 2025 .	COP-17 in 2025 extended the work toward consideration at OEWG-15 in 2026 and possible adoption at COP-18 in 2027 .

Note: The table focuses on regulatory developments that define the 2022-2024 regulatory-signaling period for EU-origin black-mass-related trade. Two additional developments are relevant but are treated differently in the analysis. First, the Proposed End-of-Life Vehicles Regulation, COM(2023) 451, introduced in July 2023, is included as a policy signal rather than binding law because it remained a proposal during the study period. It is relevant insofar as electric-vehicle batteries are a major future source of end-of-life battery material and black-mass feedstock, and the proposal addresses vehicle circularity, end-of-life treatment, traceability, and battery removal obligations. Second, Commission Delegated Decision (EU) 2025/934, adopted after the 2022-2024 observation window, updates the European List of Waste for battery-related waste and explicitly addresses intermediate fractions from battery treatment, including black mass. Because it falls outside the study period, it is not used to define the empirical window; instead, it is treated as post-period confirmation that black mass and related intermediate recycling fractions were becoming more formally recognized in EU waste classification.